Chinmay Burgul

cmburgul@gmail.com | cmburgul.github.io | linkedin.com/in/chinmay-burgul | github.com/cmburgul

Summary

Robotics Engineer with 5+ years of experience building and deploying multi-modal SLAM software stacks on real-world robots.

Technical Skills

- **Domain Expertise:** Visual-Inertial & LiDAR-Inertial SLAM, Sensor Fusion & Calibration, Kalman Filtering, Nonlinear Least-Squares Optimization & Computer Vision
- **Programming & Frameworks:** C++, Python, ROS1, ROS2, Docker, Open3D
- **Hardware Platforms:** Nvidia Jetson platforms, Aqua - Autonomous Underwater Vehicles (AUVs), Quadrupedal Robots (Ghost Vision 60, Jueying Lite2)

Education

- **University of Delaware** — Doctoral Student, Robotics — Newark, DE | 2021 – Present
- **Worcester Polytechnic Institute** — M.S. Robotics — Worcester, MA | 2018 – 2020
- **SRM Institute of Science and Technology** — B.Tech Mechatronics — Chennai, India | 2012 – 2016

Experience

- **Graduate Research Assistant** — *Jan 2021 – Present*
  University of Delaware — Newark, DE
  - Designed and deployed multi-sensor fusion SLAM frameworks integrating LiDAR, 3D sonar, vision, IMU, DVL, and joint encoders for legged and underwater robotic platforms
  - Implemented real-time C++/ROS autonomy stacks optimized for onboard compute, including state estimation, online mapping (ESDF), and motion planning on resource-constrained hardware
- **Graduate Research Assistant** — *Aug 2019 – Aug 2020*
  Worcester Polytechnic Institute — Worcester, MA
  - Formulated reinforcement learning-based controllers for dynamic within-hand manipulation (WiHM), advancing dexterous robotic grasping research
- **Machine Learning Intern** — *May 2019 – Aug 2019*
  Phood LLC — Boston, MA
  - Built an end-to-end ML pipeline for object detection model training and deployed scalable inference APIs on AWS
- **Mechatronics Engineer** — *Jan 2016 – Feb 2017*
  Entra Mechatronics Pte Lte — Chennai, India
  - Designed, prototyped, and tested a custom spherical-coordinate robotic arm optimized for automated task execution

Selected Projects

- **Underwater SLAM with the First Compact 3D Sonar**
  C++, ROS1/2, Open3D — Aqua UUV
  - Developed a visual–3D sonar extrinsic calibration methodology and characterized sensor response across diverse materials and environments
  - Built a robust multi-sensor SLAM system fusing visual, inertial, DVL, pressure, and 3D sonar measurements for high-fidelity underwater localization and mapping
- **Online Kinematic Calibration for Legged Robots**
  C++, ROS1/2 — Deep Robotics Jueying Lite2 & Ghost Vision 60
  - Developed a real-time state estimation framework for quadrupeds by tightly fusing visual, inertial, and leg kinematic measurements
  - Introduced an online method for calibrating leg link lengths and IMU–leg spatio-temporal offsets; validated performance gains on two distinct quadrupedal platforms

- **Efficient Multimodal Quadrupedal Autonomy: Algorithm, System & Application**
  C++, ROS1/2, Nvidia ORIN — Ghost Vision 60
  - Developed a full autonomy stack integrating multimodal state estimation with an online ESDF mapping system that efficiently fuses LiDAR and RGB-D data
  - Demonstrated autonomous large-scale indoor exploration by deploying the planning and mapping stack directly onto the Ghost Vision 60's onboard compute

OTHER PROJECTS

- Multi-Session Visual-Inertial Mapping; Deep RL for Within-Hand Manipulation (WiHM); Online Voxel & SDF Mapping; Image Detection & Instance Segmentation

PUBLICATIONS

C = CONFERENCE   S = IN SUBMISSION

[C.1] C. Burgul et al. (2024). **Online Determination of Legged Kinematics**. In *IEEE/RSJ IROS 2024*, pp. 9043–9049. DOI: 10.1109/IROS58592.2024.10801595

[C.2] C. Burgul & Y. Huang et al. (2026). **Underwater Dense Mapping with the First Compact 3D Sonar**. *Accepted, IEEE ICRA 2026*.

[C.3] Co-authored (2026). **Mapping Pamir: Multi-Session Visual/Inertial SLAM & 3D Reconstruction of an Underwater Shipwreck**. *Accepted, IEEE ICRA 2026*.

[S.1] C. Burgul et al. (2026). **SVIn+: Multi-Sensor SLAM with 3D Sonar, DVL, Pressure, and Visual-Inertial Sensors for Underwater Robotics**. *In submission*.

SERVICE & LEADERSHIP

- **Peer Review:** ICRA, IROS, RA-L, T-RO (2022–2026)
- **Vice-President,** Robotics Graduate Student Organization (RGSO) 2024 – 2025, University of Delaware
- **Treasurer,** Mechanical Engineering Graduate Student Organization (MEGA) 2022 – 2023, University of Delaware